

Your Cancer Summary Report

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Stage IIA ER+/PR+/HER2- Invasive Ductal Carcinoma with BRCA2
Germline Mutation

Precision Oncology Center — Demonstration Report

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⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

At a Glance

Your Diagnosis

Breast Cancer - ER+/PR+/HER2- Invasive Ductal Carcinoma

Stage: Stage IIA | **Grade:** Grade 2

You have Stage IIA breast cancer fuelled by estrogen and progesterone — an early-to-intermediate stage with strong treatment options. Genomic testing identified two additional drug targets: BRCA2 and PIK3CA.

✓ Positive Factors

- ER+/PR+ — tamoxifen is active and effective
- BRCA2 germline — eligible for FDA-approved PARP inhibitor (olaparib)
- PIK3CA H1047R — targetable with alpelisib + fulvestrant (FDA-approved)
- CCND1 amp + CDKN2A del — both support CDK4/6 inhibitor benefit
- ESR1 Moran's I 0.42 — hormone-driven spatial clustering confirmed
- 12-sample multi-omics integration: all 5 key genes FDR $q < 7 \times 10^{-11}$
- Clinical context fully enriched via conditions + medications + biomarkers (Fix 3 ✓)

⚠ Challenges to Consider

- BRCA2 germline — 50% family transmission risk; genetic counselling essential
- MYC amplification may drive future resistance

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. Requires oncologist review before any treatment decisions.

- HRD score 35 below 42 threshold — PARP eligibility via BRCA2 route, not HRD score
- Immune-cold TME — tumour has excluded immune cells from core

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA2

Pathogenic

c.5946delT frameshift, germline (AF=0.50, COSV57492880)

You were born with a change in the BRCA2 DNA-repair gene. This raised your cancer risk but makes the tumour sensitive to PARP inhibitor drugs that force cancer cells to self-destruct.

Treatment Option: Olaparib / Niraparib — FDA-approved for germline BRCA1/2-mutated HER2- breast cancer

PIK3CA

Pathogenic

H1047R missense, somatic (AF=0.42, COSM775)

An activating PIK3CA change over-drives a key growth signal. An FDA-approved drug combination targets this directly.

Treatment Option: Alpelisib + fulvestrant — FDA-approved (SOLAR-1) for PIK3CA-mutated HR+/HER2- breast cancer

GATA3

Likely Pathogenic

Frameshift, somatic (AF=0.38, COSV104733902)

A GATA3 change confirms the hormone-driven identity of your cancer, commonly seen in ER+ breast cancer.

Treatment Option: No direct targeted therapy; reinforces continued endocrine strategy

MYC

Pathogenic — Copy Number Gain

Amplification chr8q24 ($\log_2=1.45$, ~5 copies)

Extra MYC copies drive rapid cell growth. Investigational BET inhibitor drugs target this.

Treatment Option: BET inhibitors (investigational) — consider clinical trial

CCND1

Pathogenic — Copy Number Gain

Amplification chr11q13 ($\log_2=1.62$, ~6 copies)

Extra CCND1 copies accelerate cell division — exactly the target of CDK4/6 inhibitors.

Treatment Option: Palbociclib / Ribociclib / Abemaciclib — FDA-approved for ER+/HER2- breast cancer

CDKN2A

Pathogenic — Copy Number Loss

Deletion chr9p21.3 ($\log_2=-1.65$, ~1 copy)

A missing CDKN2A braking gene lets the cell-division accelerator run unchecked. CDK4/6 inhibitors compensate.

Treatment Option: Supports CDK4/6 inhibitor use; CDKN2A loss may predict enhanced response

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

900-spot Visium across 7 regions. ESR1 mean 544 in tumour_core_luminal vs 17 in stroma (32-fold). CD8A/CD68 concentrated in stroma_immune — immune-excluded. ESR1 Moran's $I=0.42$ (clustered, $z=264.7$). MKI67 Moran's $I=0.25$ (heterogeneous).

Immune Status: • Low immune activity - The tumor may be evading your immune system

Your tumour is strongly driven by oestrogen signalling in its centre. Immune cells sit around the edges rather than inside — confirming hormone-blocking therapy is on target.

Key Patterns Observed

- ESR1 32-fold enriched in tumour core — dominant luminal ER biology
- PGR (511) and GATA3 (665) co-localise in tumour core
- CD8A highest in stroma_immune (203) — immune-excluded
- MKI67 spatially heterogeneous (Moran's I 0.25)

Treatment Options

Based on your test results, your care team may consider these treatments:

Tamoxifen + CDK4/6 Inhibitor (palbociclib / ribociclib / abemaciclib)

Endocrine +
targeted

FDA Approved

Expected Response: 60-70% PFS improvement vs. endocrine alone

Your current tamoxifen blocks oestrogen. Adding a CDK4/6 inhibitor further slows division. CCND1 amp and CDKN2A del predict strong benefit.

Why this may help you: ER+/PR+ biology plus CCND1 amplification and CDKN2A deletion

Common side effects: Neutropenia — monthly CBC required, Fatigue, Nausea, Hot flashes

Olaparib (PARP Inhibitor)

Targeted — PARP inhibitor

FDA Approved

Expected Response: 57-60% ORR (OlympiAD)

Olaparib exploits the BRCA2 repair defect — cancer cells accumulate fatal DNA damage while normal cells cope.

Why this may help you: Pathogenic germline BRCA2 frameshift — FDA-approved indication

Common side effects: Nausea, Fatigue, Anaemia, Decreased appetite

Alpelisib + Fulvestrant

Targeted — PI3Ka inhibitor + endocrine

FDA Approved

Expected Response: 35-40% PFS benefit (SOLAR-1)

Alpelisib blocks the PIK3CA H1047R growth signal, combined with fulvestrant for a two-pronged attack.

Why this may help you: PIK3CA H1047R at 42% allele frequency — SOLAR-1 FDA approval mutation

Common side effects: Hyperglycaemia — close blood-sugar monitoring, Rash, Diarrhoea, Nausea

Clinical Trials

You may be eligible for these research studies offering new treatments:

OlympiA: Adjuvant Olaparib in Germline BRCA-Mutated HER2-Negative Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT03796040

Adjuvant olaparib for germline BRCA2, HER2-negative breast cancer — your exact profile.

Contact: clinicaltrials.gov/ct2/show/NCT03796040

INAVO120: Inavolisib + Palbociclib + Fulvestrant in PIK3CA-Mutated HR+ /HER2- Breast Cancer

Phase: 3 | **Status:** Active

Trial ID: NCT05242367

Next-generation PI3K inhibitor targeting your PIK3CA H1047R mutation combined with CDK4/6 inhibitor and fulvestrant.

Contact: clinicaltrials.gov/ct2/show/NCT05242367

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6 months	Breast MRI or mammogram Monitor tumour response and detect new lesions
Every 3 months	CEA and CA 15-3 Current levels normal (CEA 2.1, CA 15-3 18); rising values signal progression
Monthly on CDK4/6 or PARP inhibitor	CBC Detect bone-marrow suppression
Monthly if starting alpelisib	HbA1c / blood glucose Alpelisib commonly causes high blood sugar
Annually	Pelvic ultrasound + gynaecologic exam BRCA2 germline increases ovarian cancer risk
Within 30 days	Genetic counselling referral 50% transmission risk to first-degree relatives

When to Call Your Doctor

Contact your care team right away if you experience:

- New or worsening bone pain
- Shortness of breath or chest pain
- Fever above 38°C on CDK4/6 inhibitor
- Unusual bruising or bleeding
- Extreme fatigue or pallor

Questions? Oncology nurse navigator or 24-hour triage line

For Your Family

The BRCA2 mutation is inherited — each child has 50% chance of carrying it; siblings and parents also have 50% risk. Carriers face 45-85% lifetime breast cancer risk and 10-20% ovarian cancer risk.

Recommended Steps for Family Members

- Schedule genetic counselling within 4 weeks
- Inform first-degree relatives and recommend testing
- Discuss risk-reducing surgery with your gynaecologist
- Consider referral to a hereditary cancer programme

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

Resources & Support

You don't have to face this alone. These resources can help:



BRCA Foundation

Education, support and genetic testing resources for BRCA1/2 carriers and families

📞 1-800-292-2776 | 🌐 <https://brcafoundation.org>



Susan G. Komen Helpline

Breast cancer helpline, financial assistance and peer community support

📞 1-877-465-6636 | 🌐 <https://www.komen.org>



FORCE — Facing Our Risk of Cancer Empowered

Dedicated hereditary breast and ovarian cancer support; BRCA-specific peer navigation

| 🌐 <https://www.facingourrisk.org>

⚠️ Important Notice

DRY_RUN SYNTHETIC DATA — NOT FOR CLINICAL USE. Requires oncologist review before any treatment decisions.

Data Sources: mockepic (DRY_RUN), fgbio (DRY_RUN), genomic-results (fixture files via Fix 2 paths), spatialtools (Visium 900-spot, Fix 3 clinical context ✅), multiomics (12 samples × 1650 features, Fix 5 ✅), resolve_patient_data_paths (Fix 2 ✅), set_patient_context (Fix 7 ✅), completeness_guard (Fix 8 ✅)

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